



ESCUELA DE INGENIERÍA
FACULTAD DE INGENIERÍA

Lecture 24: Oligopoly III

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ICS2523 Microeconomics

Lecture 24

Last lecture, we:

- Reviewed examples of matrix games
- Saw an example of continuum of actions → Tragedy of the Commons
- Started with Oligopoly models: Bertrand, Cournot

In today's lecture, we will finish studying **Oligopoly**:

- Oligopoly models: Cartel, Stackelberg
- Analyse oligopolies and repeated games

“LO QUE PASA EN LA SALA
NO QUEDA EN LA SALA
QUEDA EN LA
ENCUESTA DOCENTE”

Encuesta Docente
en Canvas

Desde el 11 de junio



y eso lo
cambia todo



PONTIFICIA
UNIVERSIDAD
CATÓLICA
DE CHILE

Oligopoly models

Market Failures

Perfect Cartel

Market Failures

11. Oligopoly Perfect Cartel

Cartel's objective is to **maximize joint profit**:

$$\text{Max} \sum_{j=1}^n \pi_j = P(Q) \sum_{j=1}^n q_j - \sum_{j=1}^n C_j(q_j)$$

Taking the first-order condition with respect to q_i gives

$$\frac{\partial \left(\sum_{j=1}^n \pi_j \right)}{\partial q_i} = P(Q) + P'(Q) \sum_{j=1}^n q_j - C_i'(q_i) = 0$$

$$\frac{\partial \left(\sum_{j=1}^n \pi_j \right)}{\partial q_i} = P(Q) + P'(Q)Q - C_i'(q_i) = 0$$

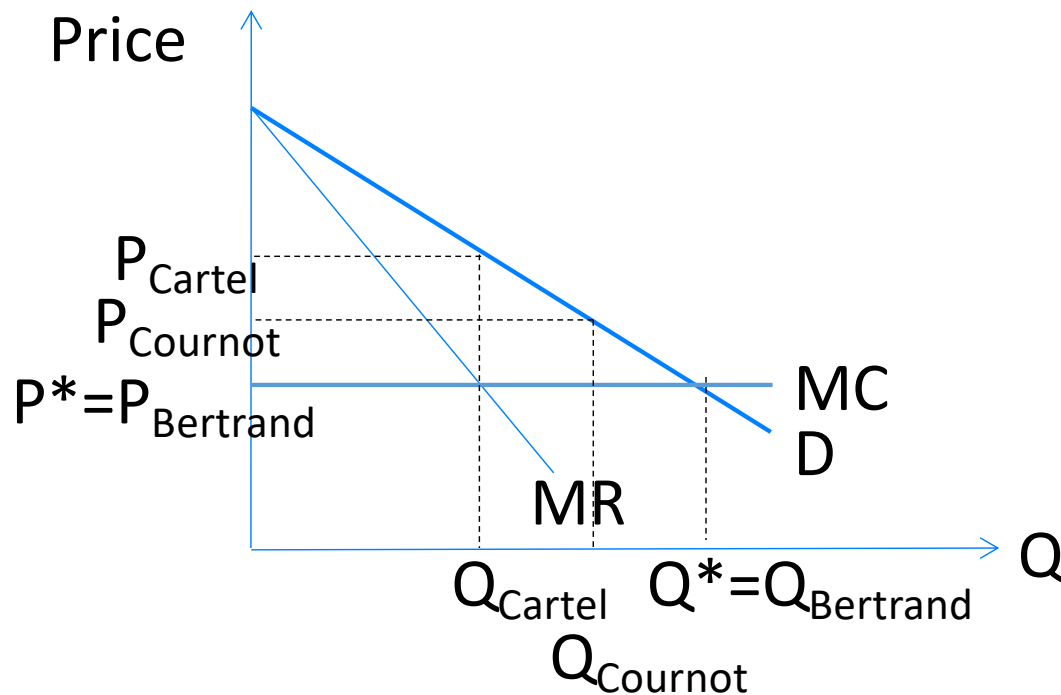
11. Oligopoly Perfect Cartel

In this case, term $P'(Q)Q$ is larger than with Cournot firms $(P'(Q)q_i)$

→ Cartel accounts for the fact that an additional unit of firm's i output reduces the revenue earned on ALL firms' existing output (by reducing market price)

Cournot firms will be overproducing relative to the cartel, which is good for society (more total welfare in the case of Cournot than the cartel).

Oligopoly: Comparison between Cournot, Bertrand and perfect Cartel



Activity 1

Duopoly example

Two firms (each owns a spring with a large supply of water). Producers must decide how much to provide the market with the following cost of pumping and bottling q_i liters:

$$C_i(q_i) = cq_i$$

Inverse demand for spring water is: $P(Q) = a - Q$, where $Q = q_1 + q_2$

Find the price, quantities offered and level of profit for each firm in a:

- a) Bertrand model
- b) Cournot model
- c) Perfect cartel

11. Oligopoly Perfect Cartel

Cartel instability

For each firm belonging to the cartel, it is true that:

$$\pi_i^{Cartel} > \pi_i^{Cournot}$$

(each one is better off colluding/being cooperative than no cooperative).

Nevertheless, it is **very tempting** to deviate → cartels are unstable (and illegal)

Cartel instability

From Activity 3, if firm 1 assumes that firm 2 will produce the quantity agreed by the cartel ($q_2 = q_2^c = \frac{(a-c)}{4}$), which should be the optimum response of firm 1 (q_1^{**})?

Cartel instability

Firm 1 will:

$$\begin{aligned} \text{Max}_{q_1} \pi_1(q_1) &= P(q_1 + q_2) \cdot q_1 - c \cdot q_1 \\ &= (a - q_1 - ((a - c)/4)) \cdot q_1 - c \cdot q_1 \\ &= \left(\frac{3}{4}a - \frac{3}{4}c - q_1 \right) \cdot q_1 \end{aligned}$$

First order condition:

$$\begin{aligned} \left(\frac{3}{4}a - \frac{3}{4}c - q_1^{**} \right) - q_1^{**} &= 0 \\ q_1^{**} &= \frac{3}{8}(a - c) \end{aligned}$$

So, if firm 1 deviates, its profit will be:

$$\pi_1^{**} = P(q_1^{**} + q_2^c) \cdot q_1^{**} - c \cdot q_1^{**} = \frac{9}{64}(a - c)^2$$

Cartel instability

$$\pi_1^{**} = \frac{9}{64}(a - c)^2 > \frac{1}{8}(a - c)^2 = \pi_1^{cartel}$$

Firm 2, complying with the original agreement, producing $(a-c)/4$ will get a profit:

$$\pi_2^- = P(q_1^{**} + q_2^c) \cdot q_2^c - c \cdot q_2^c = \frac{6}{64}(a - c)^2$$
$$\pi_2^- = \frac{6}{64}(a - c)^2 < \frac{1}{8}(a - c)^2 = \pi_2^{cartel}$$

11. Oligopoly Perfect Cartel

Cartel instability

Therefore, firms in a cartel face a similar situation as the Prisoner's Dilemma:

		Firm 2	
		Coordinated	Not coordinated
Firm 1	Coordinated	$\pi_1^{cartel}, \pi_2^{cartel}$	π_1^-, π_2^{**}
	Not coordinated	π_1^{**}, π_2^-	$\pi_1^{cournot}, \pi_2^{cournot}$

Stackelberg Model

Market Failures

12. Oligopoly: Stackelberg Model

Similar to a duopoly version of the Cournot model, except that rather than simultaneously choosing the quantities of their identical outputs, **firms move sequentially**:

- Firm 1 (the leader) chooses its output first
- Then firm 2 (follower) chooses output after observing firm 1's output

12. Oligopoly: Stackelberg Model

We will use **backward induction** to solve for the subgame-perfect equilibrium of this sequential game.

- Step 1: **Follower's output choice** → Firm 2 chooses output q_2 that maximizes his profit, taking q_1 as given → $BR_2(q_1)$ and should be equal to the result computed on Cournot's BR.
- Step 2: **Leader's output choice** → Firm 1 recognizes that it can influence the follower's actions → profit function for Firm 1 will include $BR_2(q_1)$

$$\pi_1 = P(q_1 + BR_2(q_1))q_1 - C_1(q_1)$$

12. Oligopoly: Stackelberg Model

The first-order condition for q_1 is:

$$\frac{\partial \pi_1}{\partial q_1} = P(Q) + P'(Q)q_1 + P'(Q)BR_2'(q_1)q_1 - C_1'(q_1) = 0$$

The third term accounts for the strategic effect of firm's 1 output on firm 2's $\rightarrow$ firm 1 will produce more than it would have in the Cournot model $\rightarrow$ firm 2 will reduce q_2

If the leader's output choice is not observable, we are back in the Cournot game.

12. Oligopoly: Stackelberg Model

Let's see how our example of the duopoly changes with the Stackelberg model:

Step 1: Follower's output choice $\rightarrow BR_2(q_1) = \frac{a - q_1 - c}{2}$

Step 2: Leader's output choice $\rightarrow$

$$\pi_1 = P(q_1 + BR_2(q_1))q_1 - C_1(q_1)$$

$$\pi_1 = P(q_1 + BR_2(q_1))q_1 - C_1(q_1) = \left[a - q_1 - \left(\frac{a - q_1 - c}{2} \right) - c \right] q_1$$

$$\pi_1 = \frac{1}{2}(a - q_1 - c)q_1$$

12. Oligopoly: Stackelberg Model

The first-order condition with respect to q_1 is:

$$\frac{\partial \pi_1}{\partial q_1} = \frac{1}{2}(a - 2q_1 - c) = 0$$

$$\rightarrow q_1^* = (a - c)/2 \rightarrow q_2^* = (a - c)/4$$

The Stackelberg model equilibrium involves higher output and profit for firm 1 compared to the Nash equilibrium.

Homework Lecture 24

Stackelberg model

Two identical firms (each owns a spring with a large supply of water). Producers must decide how much to provide the market with the following cost of pumping and bottling q_i liters:

$$C_i(q_i) = cq_i$$

Inverse demand for spring water is: $P(Q) = a - Q$, where $Q = q_1 + q_2$ and $a > c$

Firm 1 is the leader (decides first), and firm 2 is the follower.

Find the price, quantities offered and level of profit for each firm in a Stackelberg model.

Graph the best response functions and identify the points representing the Cournot and Stackelberg equilibria

Repeated Games

Market Failures

14. Repeated Games

So far, each player makes one choice, and the game ends. In many real-world settings, players play the same game repeatedly.

The simple constituent game that is played repeatedly is called the **stage game**. Repeated play of the stage game opens up the possibility of cooperation in the equilibrium.

Players can adopt **trigger strategies**, whereby they continue to cooperate as long as all have cooperated up to that point but revert to playing the Nash equilibrium if anyone deviates from cooperation.

14. Repeated Games

Finitely repeated games

For many stage games, repeating them a known, finite number of times does not increase the possibility for cooperation.

For instance, imagine the **Prisoners' Dilemma was played for T periods**. Use backward induction to solve for the subgame-perfect equilibrium: The lowest subgame is the Prisoners' Dilemma stage game played in period T . Regardless of what happened before, the Nash equilibrium is for both to fink.

Folding the game back to period $T-1$, we have just seen that nothing that happens in period $T-1$ affects what happens subsequently because both players fink → It is as though period $T-1$ was the last, and the Nash equilibrium is for both to fink.

14. Repeated Games

Finely repeated games

As shown by Reinhard Selten, for any stage game with a unique Nash equilibrium, the unique subgame-perfect equilibrium of the finitely repeated game involves playing the Nash equilibrium every period.

If the stage game has multiple Nash equilibria, it may be possible to achieve some cooperation in a finitely repeated game. Players can use trigger strategies, sustaining cooperation in early periods on an outcome that is not an equilibrium of the stage game, by threatening to play in later periods the Nash equilibrium that yields worse outcome for the player who deviates from cooperation.

14. Repeated Games

Infinitely repeated games

With infinitely repeated games there is no definite ending period T from which to start backward induction.

One difficulty with infinitely repeated games involves adding up payoffs across periods. An infinite stream of low payoffs sum sums to infinity just as an infinite stream of high payoffs.

How can the two streams be ranked?

Discounting

14. Repeated Games

Infinitely repeated games

Let δ be the discount factor measuring how much a payoff unit is worth if received one period in the future rather than today (δ is inversely related to the interest rate and can also incorporate uncertainty about whether the game continues in future periods, $\delta < 1$).

Present value of \$1 payment per period from now on is:

$$\$1 + \$1 \cdot \delta + \$1 \cdot \delta^2 + \$1 \cdot \delta^3 + \dots = \sum_{i=0}^{\infty} \$1 \cdot \delta^i = \frac{\$1}{1 - \delta}$$

Factoring in a probability that the repeated game ends after each period makes the setting of an infinitely repeated game more believable. The **crucial issue of an infinitely repeated game** is not that it goes on forever, but that **its end is indeterminate**.

14. Repeated Games

Infinite geometric series (brief recap)

For $\delta < 1$:

$$a + a \cdot \delta + a \cdot \delta^2 + a \cdot \delta^3 + \dots = \sum_{i=0}^{\infty} a \cdot \delta^i = \frac{a}{1 - \delta}$$

$$\sum_{i=0}^{\infty} a \cdot \delta^i - \delta \sum_{i=0}^{\infty} a \cdot \delta^i = a$$

$$\sum_{i=0}^{\infty} a \cdot \delta^i = \frac{a}{1 - \delta}$$

14. Repeated Games

Infinitely repeated games

Suppose we expect two neighboring gasoline stations to play a pricing game each day, until electric cars replace gasoline-powered ones. Gasoline stations will be uncertain about the end of gasoline-powered cars → end of their pricing game is indeterminate

Trigger strategies have players cooperating with each other as long as no one has deviated. For a trigger strategy to work the punishment must be severe enough to deter deviation in the first place.

14. Repeated Games

Infinitely repeated games

Let's now imagine a Prisoners' Dilemma game:

		Suspect 2	
		Fink	Silent
Suspect 1	Fink	$u_1=1, u_2=1$	$u_1=3, u_2=0$
	Silent	$u_1=0, u_2=3$	$u_1=2, u_2=2$

Let's now imagine a Prisoners' Dilemma game and suppose that both players are using a specific trigger strategy: continue being silent if no one has deviated, fink forever if anyone has deviated to fink in the past.

14. Repeated Games

Infinitely repeated games

To show that this trigger strategy forms a subgame-perfect equilibrium, we need to check that a player cannot gain from deviating.

Along the equilibrium path both players are silent every period, this provides each with a payoff of 2 every period for a present discounted value of:

$$\begin{aligned} V^{eq} &= 2 + 2\delta + 2\delta^2 + 2\delta^3 + \dots \\ &= 2(1 + \delta + \delta^2 + \delta^3 + \dots) = \frac{2}{1 - \delta} \end{aligned}$$

14. Repeated Games

Infinitely repeated games

A player who deviates by finking, earns 3 in that period, but then both players fink every period from then on (each earning 1 per period):

$$\begin{aligned} V^{dev} &= 3 + 1 \cdot \delta + 1 \cdot \delta^2 + 1 \cdot \delta^3 + \dots \\ &= 3 + \delta(1 + \delta + \delta^2 + \delta^3 + \dots) = 3 + \frac{\delta}{1 - \delta} \end{aligned}$$

The trigger strategies for a subgame-perfect equilibrium if

$$\begin{aligned} V^{eq} &\geq V^{dev} \\ \frac{2}{1 - \delta} &\geq 3 + \frac{\delta}{1 - \delta} \rightarrow \delta \geq \frac{1}{2} \end{aligned}$$

14. Repeated Games

Infinitely repeated games

Thus, players will find continued cooperative play desirable provided they do not discount future gains from such cooperation too highly.

If $\delta < \frac{1}{2}$, then no cooperation is possible in this game.

The discount factor (δ) is crucial in determining whether trigger strategies can sustain cooperation in any stage game.

Example: Repeated games

Suppose only two firms produce a certain medical device used in surgery. The medical device is produced at a constant average and marginal cost of \$10, and the demand for the device is given by

$$Q = 5000 - 100P$$

- a) If a Bertrand game is played in a single period, find each firm's equilibrium quantities, price and profits.
- b) If there was a monopoly, how many devices would be sold, at what price and how much will the profits be?
- c) Find the condition for the discount factor (δ) so that collusion is attractive in an infinitely repeated game. In this case, they will use the monopoly price in the stage game, and each firm will earn an equal share of the monopoly profit.

Example: Repeated games

- a) If a Bertrand game is played in a single period, find the equilibrium quantities, price and profits of each firm.

If a Bertrand game is played in a single period, then each firm will charge $P=MC=\$10 \rightarrow$ acknowledging this price in the demand for the device:

$$Q = 5000 - 100P$$
$$Q = 5000 - 100 \cdot 10 = 4000$$

$Q = 4,000$ devices will be sold. Profits are equal to zero.

Example: Repeated games

- b) If there was a monopoly, how many devices would be sold, at what price and how much will the profits be?

In a monopoly, the firm maximizes its profits knowing that it is the only firm facing the total demand $\rightarrow MR=MC$

$$\pi = P \cdot Q - C(Q) = \left(50 - \frac{Q}{100}\right) \cdot Q - 10Q$$
$$\frac{\partial \pi}{\partial Q} = -\frac{Q}{100} + 50 - \frac{Q}{100} - 10 = 0$$

$$Q_m = 2,000$$

$$P_m = \$30$$

$$\pi = \$40,000$$

Example: Repeated games

- c) Find the condition for the discount factor (δ) so that collusion is attractive in an infinitely repeated game. In this case, they will use the monopoly price in the stage game, and each firm will earn an equal share of the monopoly profit.

Since the monopoly price is \$30 and the total profit is \$40,000, firms have a clear incentive to consider collusive strategies.

The firms may collude tacitly to sustain the monopoly price (each firm earning an equal share of the monopoly profit) and use the **grim trigger strategy** of continuing to collude as long as no firm has undercut the monopoly price in the past. They will revert to the stage-game Nash equilibrium of marginal cost pricing every period from then on if any firm deviates by undercutting.

Example: Repeated games

Successful tacit collusion provides the following profit stream for one of the firms:

$$V^{collude} = \frac{\pi_m}{2} + \delta \cdot \frac{\pi_m}{2} + \delta^2 \cdot \frac{\pi_m}{2} + \dots$$

$$V^{collude} = \frac{\pi_m}{2} (1 + \delta + \delta^2 + \dots)$$

$$V^{collude} = \frac{\pi_m}{2} \cdot \frac{1}{1 - \delta} = 20,000 \cdot \frac{1}{1 - \delta}$$

Example: Repeated games

When undercutting the Monopoly price **SLIGHTLY** a firm can obtain essentially all the monopoly profit for itself in the current period. This deviation would trigger the grim strategy punishment of marginal cost pricing in the second and future periods, so all firms will earn zero profits from there on:

$$V^{deviate} = \pi_m + 0 + 0 + \dots = \$40,000$$

For this deviation not to be profitable, we must have:

$$V^{deviate} \leq V^{collude}$$

$$40,000 \leq 20,000 \cdot \frac{1}{1 - \delta}, \delta \geq 1/2$$

Individual Activity

On a piece of paper, write down **2-3 concepts** that you learnt today. Please write down the concept and what you learnt about it in your own words.

If you have any **questions** about what we discussed today, write them down too. We Will go through them at the beginning of the next class.

Finally, if there is a word you didn't understand, write it down so it can be part of the **glossary**.